Fiscal Year:	FY 2012	**Task Last Updated:**	FY 07/26/2012
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Project Title:	Dynamical Selection of Three-Dimensional Interfacial Patterns in Directional Solidification		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2007	**End Date:**	11/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 11/30/2012 (Ed., 6/22/2012) NOTE: End date changed to 5/31/2012 (from 5/31/2010) per NSSC info dtd 9/2009 (10/2009) NOTE: Received NCE to 5/31/2010 (from 5/31/2009) per NSSC info dtd 6/2009 (10/2009)		
Key Personnel Changes/Previous PI:	Dr. Jiho Gu left the research group at the end of 2011.		
COI Name (Institution):	Karma, Alain (Northeastern University)		
Grant/Contract No.:	NNX07AK69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed research consists of experiments on the International Space Station on the dynamics of three-dimensional interface pattern evolution in directional solidification, and the supporting critical ground-based research and the rigorous modeling to analyze observations from the low gravity experiments. Low gravity experiments were successful in that no effect of convection was observed. Initial results from the space station were analyzed to characterize the dynamical evolution of spatially extended patterns. Several new results were observed that include the continuous variation in primary spacing with velocity over the cell-dendrite growth regime, and first observations of the formation of multiplets in three-dimensions and the presence of novel oscillating modes during the growth of cellular array at low velocities. A rigorous theoretical model, using the phase-field approach was developed that gave excellent agreement with the experimental observation on the complex oscillating modes observed experimentally. Detailed experimental studies were also carried out terrestrially in a thin sample in which diffusive growth conditions are present to examine the variation in primary spacing with velocity and to establish some critical characteristics of multiplets observed in low gravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of fundamental quantitative knowledge of microstructure evolution under dynamic growth conditions is crucially important for improving large scale industrial processes from casting to welding. The results will provide basis to predict and design reliable processing conditions that optimize microstructures and hence material properties and performance. These results are crucial in the design of high precision castings for aerospace and defense industries. This research provides for the first time the benchmark results on three-dimensional spatial arrangement of cells and dendrites since convection effects strongly influence the patterns and prevent the fundamental physics that govern these patterns under diffusive growth conditions. The benchmark data obtained under low gravity conditions will be a cornerstone for the validation of phase-field approach through unique quantitative comparison of model predictions and benchmark experiments in three dimensions. These models can then be used successfully in the design of casting parameters that are crucial for the production of high quality precision castings by U.S. industries. The knowledge gained from this study will also be applied to other pattern forming phenomena that occur in different fields of science such as combustion, fluid physics, cryo-biology, biology, and food science.
	In many commercial processing technologies, such as casting, welding, and brazing, the conditions under which the solidification process occurs govern the resulting microstructure that has a profound effect on the mechanical properties and reliability of a wide range of commercial metallic products of vital importance for the automotive, aerospace, and other industries. In most precision casting and welding processes, solidification occurs with a cellular or dendritic interface, and one of the critical parameters relevant to these technologies is the spatial arrangement of cells and dendrites in three-dimensions that influence the solute segregation patterns in the intercellular and interdendritic region. From a fundamental standpoint, dendritic microstructure development is controlled by a delicate interplay between anisotropic capillary forces acting on the nanometer scale of the solid-liquid interface and macroscopic transport of heat and solute on much larger scales, which is strongly influenced by convection under terrestrial conditions. Although the importance of dendritic structures in controlling the properties and reliability of castings has been recognized for over two centuries, convection effects are highly complex and mask the fundamental physics of the complex evolution of solidification structures. Directional solidification experiments under microgravity conditions thus provide a unique setting to investigate spatially extended three-dimensional arrays of cells and dendrites under diffusive growth conditions, and we have performed such studies for the first time on the International Space Station, which have provided benchmark results for developing and testing theoretical models that can be used to optimize solidification conditions during the high precision castings of interest to the U.S. industries. During this reporting year we have carried out experiments on the International Space Station and we shall describe some of these results first, and then show the development of a theoretical model to analyze and predict these experimental observations. Finally, we shall describe some of the ground-based studies in thin sample in which solidification occurs under diffusive growth conditions, but the array formation is restricted to a single row of cells or dendrites. I. Results from experiment on International Space Station We shall first emphasize the success of these experiments. Under terrestrial conditions, convection effects cause a significant variation in microstructure from the center to the edge of the sample, with cellular spacing decreasing towards the wall and a planar front stabilizes near the wall. Our experimental results under microgravity conditions, however, show a uniform microstructure along the cross-section that indicates that convections effects are not present, and the spatio-temporal evolution of the three-dimensional solidification structure occurs under diffusive growth conditions. Furthermore, the sample maintained its integrity over a long period of time during which experiments were conducted, and no bubble formation was observed. All components of the hardware, such as the translation stage, thermal profile, and optical and laser interference imaging worked very well. Experiments under low gravity conditions have been carried out on the International Space Station using the Directional Solidification Insert (DSI) in the DECLIC facility. Succinonitrile-0.24 wt% camphor alloy was used in this study since it is transparent so that the dynamics of three-dimensional interface pattern evolution under different growth conditions could be visualized in situ, and the images of the interface were captured by two cameras that viewed the interface pattern from the side and from the top of the sample. In addition, a laser interference technique was used to reconstruct the three-dimensional shapes of cells and dendrites near the tip region. A total of 29 experiments were carried out over 100 days of experiments, and some images were down loaded in real time, and all images were recorded on a disc on the International Space Station that were recently provided to the scientific team. . Experiments were carried out over a range of velocities to obtain cellular and dendritic arrays, and to examine the conditions under which cellular to dendritic transition occurs. The time evolution of the interface pattern is characterized from the initial planar interface, to the onset of instabilities, dynamics of spacing adjustment and the final spatial arrangement. It is found that the final spatial arrangement is close to a hexagonal arrangement with some disorder. However, more complex arrangements in which both hexagonal and square array are present in the same experiment are also observed, and these are correlated with the presence of a small angle boundary in the initial solid seed crystal. The time evolution of primary spacing is characterized by measuring the distances between the centers of the cells, and a distribution curve is obtained. The deviation from a perfect hexagonal arrangement is also determined by using the minimum spanning tree approach that gives the average distance and the deviation from the perfect hexagonal array. The distribution in spacing implies that a range of spacing is stable under given growth conditions.

Task Progress:	The variation the average primary spacing with velocity was characterized along with the distribution in spacing at each velocity. The spacing was found to decrease continuously with velocity and no discontinuity in spacing was observed at the transition from cells to dendrites. This is in contrast to the spacing variation in a linear array grown in thin samples at 1g, where a sharp increase in spacing is observed at the cell-dendrite transition, as we will show in the section on ground-based studies. Another important observation from low gravity experiments is the formation of multiplets in three-dimensions. Although multiplets have been observed in a linear array in thin samples, no observation in three-dimensional array has been seen before in ground-based experiments. In multiplets, some cells break-up into few sub-cells with smaller spacing and smaller amplitudes. The velocity range over which these multiplets exist is now being determined. Finally, novel oscillating modes during the cellular growth at low velocity are observed in low gravity experiments that show that the oscillatory behavior is governed by the diffusive growth process. In a hexagonally arranged cellular array, at V= 1.0 μm/s, three groups of cells, each group containing three cells, show coordinated oscillations in time with periodic change in the amplitude and cross- cross- sectional area. A phase difference of 120o is observed between these three groups. A detailed theoretical analysis of this oscillatory phenomenon has been carried out and presented in the next section. The results described above are obtained only from a limited set of experiments and significantly more data need to be analyzed from all images captured in real time on ISS. II. Theoretical modeling We have carried out the first quantitative comparison of directional solidification experiments conducted on the International Space Station, where a transparent succinonitrile (SCN)-camphor organic alloy was used for in-situ visualization of the solid-liquid interface dynamics, and three-dimensional phase-field modeling of those experiments. The simulations exploit a quantitative phase-field model of the directional solidification of a dilute binary alloy and a massively parallel implementation of this model on graphics processing units (GPU). While previous phase-field simulations have been limited to two dimensions or small system sizes in three dimensions, the present GPU implementation has enabled us to simulate spatially extended systems of many cells in three dimensions on experimentally relevant length and time scales, with 1 hour of computation time corresponding approximately to 1 hour of experimental time. The phase-field simulations to date have allowed us to characterize a novel experimentally observed oscillatory mode of the cellular structure. The simulations have reproduced several salient features of this oscillatory growth mode. Firstly, the oscillations occur on a slow diffusive time scale of ~ 49 minutes that is in excellent agreement with observed period of ~ 45 minutes in the ISS experiments; microgravity played a crucial role in uncovering the existence of these oscillations that are normally altered or masked by convection effects under terrestrial conditions. Secondly, also as experimentally observed, the oscillations involve a coordinated motion of three groups of cells of the underlying hexagonal array structure that oscillate with a 1200 phase sift with respect to each other. The model has quantitatively predicted these observations. Ongoing work focuses on understanding the mechanism of those oscillations and how they are influenced by the structure of the array and growth parameters. III. Ground-based experiments in thin samples Experiments were carried out in ground-based study in thin samples where diffusive growth is obtained. Two sets of experiments were carried out: (1) to compare the results in three-dimensions obtained from the space station with the results in a thin sample, and (2) to examine the characteristics of multiplets in thin samples. The first set of experiments was carried out to examine the primary spacing variation with velocity. Experiments were carried out in thin samples of 200 μm and 100 μm thickness to examine the effect of sample thickness on cell and dendrite spacing. Two significant observations were made. (1) A jump in the spacing is observed at the cell-dendrite transition in both samples, while the spacing variation was found to show no jump in spacing in microgravity experiments in three-dimensions. (2) The spacing values for cells are significantly smaller in thin sample than in bulk samples under microgravity conditions for the sample thickness of 200 μm. However, the sample thickness was found to significantly influence the spacing in that a much larger spacing was observed when the thickness was reduced to 100 μm. Consequently, it is not possible to extend the observations in thin samples where a single array forms to those in the bulk system where a three-dimensional array forms. Consequently low gravity experiments are essential to characterize the spatial arrangement of arrays. Furthermore, experiments in thin samples cannot give the spatial arrangement, viz. hexagonal or square array that is observed in low gravity experiments. Another set of experiments was carried out in thin samples to examine the formation of multiplets. The formation of multiplet gives two characteristic spacing: one between the cells, which is large, and the other between the sub-cells, which is much smaller. When the local cell spacing is large, the cell tends to split, but the spacing is much smaller than that required for two cells or a dendrite. Thus, the array takes on a new branch of solutions in which a cell breaks up into different sub-cells. It was further observed that the duplex structure is symmetrical such that the two sub-cells were mirror images of each other. Simulations of multiplet formation in three-dimensions will be carried out to establish the important characteristics of multiplets.
Bibliography Type:	Description: (Last Updated: 04/03/2024)
Papers from Meeting Proceedings	Pont G, Barde S. Blonde D, Zappoli B, Lecoutre C, Beysens D, Hicks M, Hegde U, Hahn I, Bergeon N, Billia B, Mangelinck-Noël N, Ramirez A, Trivedi R "Declic, Soon Two Years of Successful Operations." 62nd International Astronautical Congress, IAC2011, Cape Town, South Africa, October 3-7, 2011. 62nd International Astronautical Congress, IAC2011, Cape Town, South Africa, October 3-7, 2011. Paper number IAC-11.A2.5.4.x10452 ; http://www.cnes.fr/automne_modules_files/standard/public/p8782_162077f57febf6db1584b0df03fdc051IAC_11_A2_5_4.pdf (accessed 7/26/2012) , Oct-2011